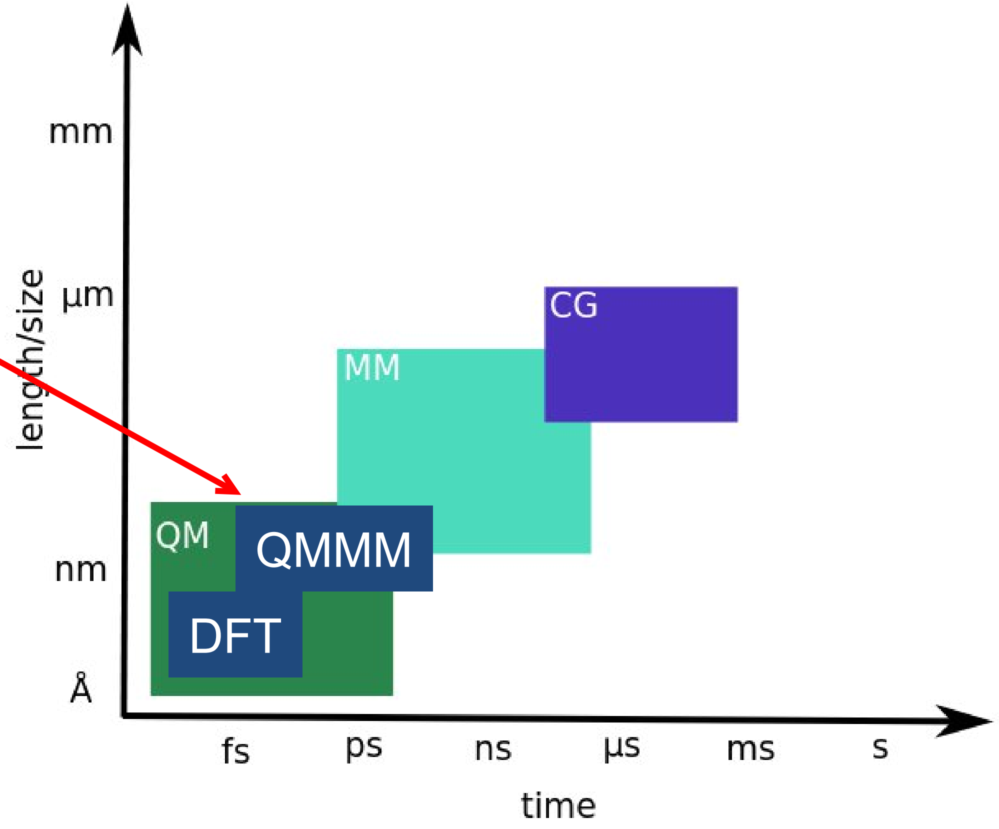
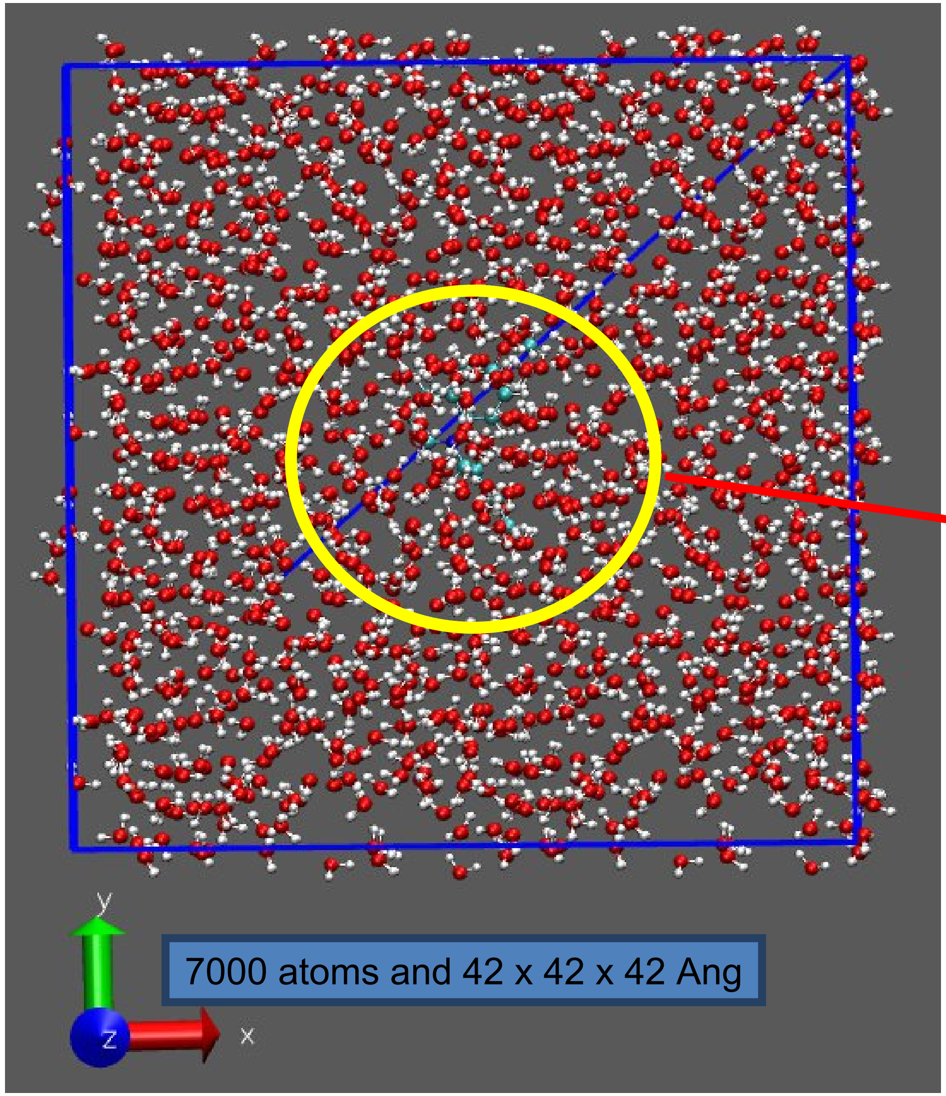
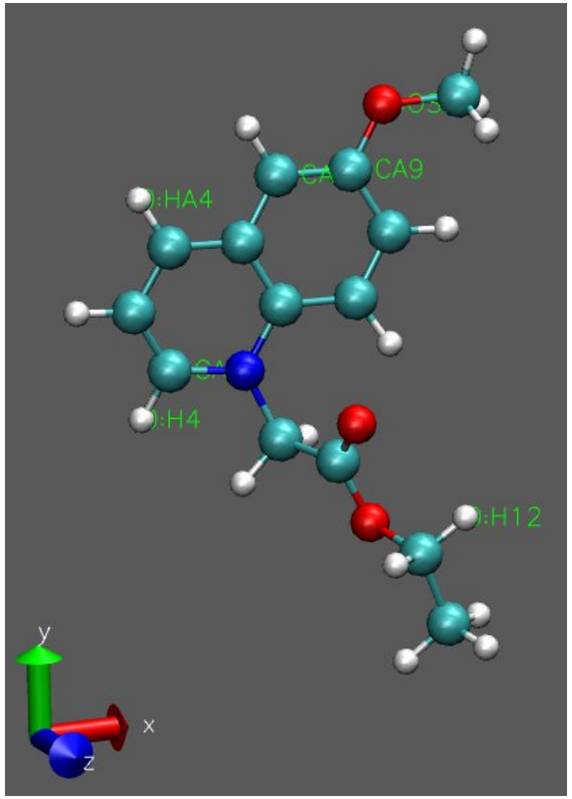
QM/MM



Why QM/MM

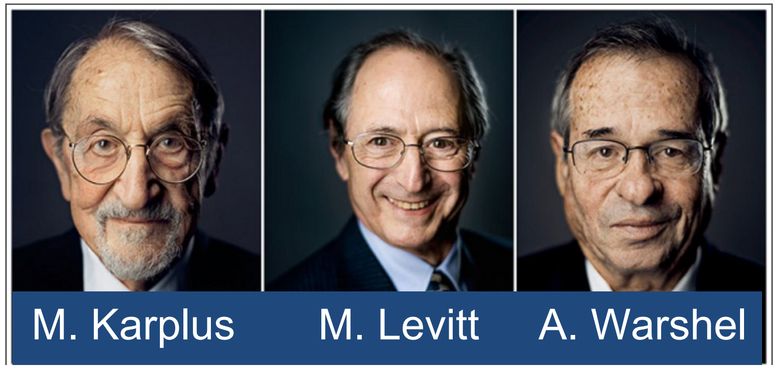


N-(6-methoxyquinolyl) acetoethyl ester

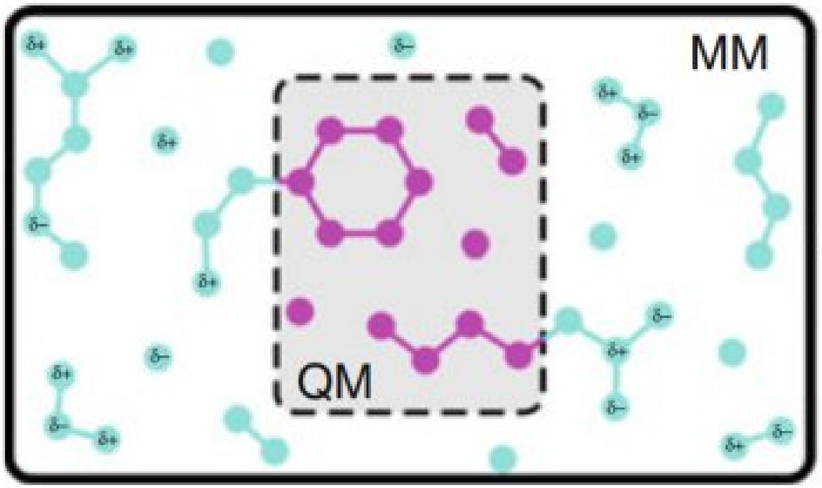


Why QM/MM

- Pure QM: expensive for 1000+ atoms
- Pure MM: not all reactions defined
- Relevant part with QM: Density Functional Theory [1]
- Rest with low level MM: Mostly All - Atoms FF [2,3]
- QM/MM Coupling [4,5]
- Advantage: Longer & accurate simulations at reduced cost [5]



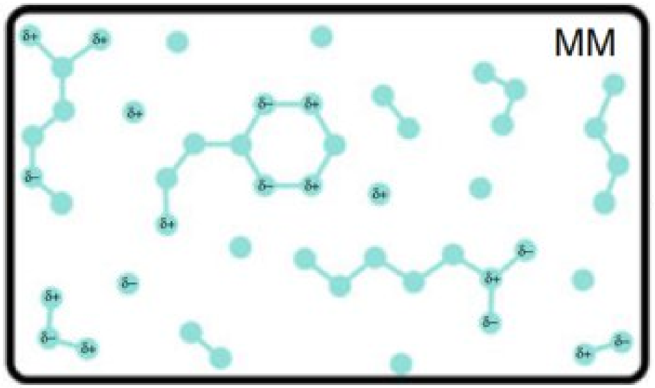
Nobel Prize 2013

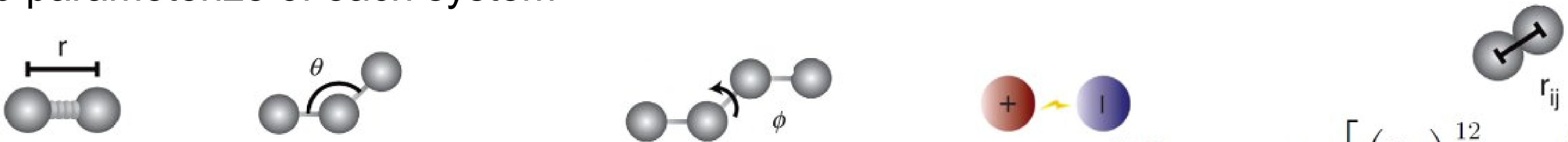


[1] J. Chem. Phys. 152, 194103 (2020)
[2] J. Phys. Chem. B 108, 1255–1266 (2004)
[3] J. Phys. Chem. 91, 6269–6271 (1987)
[4] J. Chem. Theory Comput. 1, 1176–1184 (2005)
[5] Angew. Chem. 48, 1198–1229 (2009)

Molecular Mechanics (MM)

- Functional forms and parameters define potential energy
- potentials derived from experiments or QM level simulations
- Electrons neglected in the model
- Cannot explain bond-breaking or forming, charge transfer, polarization
- need to parameterize of each system





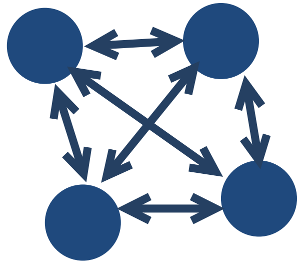
$$E_{\mathrm{system}} = \underbrace{\sum k_b (r_{ij} - r_0)^2 + \sum k_\theta (\theta_{ijk} - \theta_0)^2 + \sum k_\phi [1 + \cos(n\phi - \delta)]}_{E_{\mathrm{bonded}}} + \underbrace{\sum_{\substack{i,j \\ i \neq j}} e^2 \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \sum_{\substack{i,j \\ i \neq j}} \epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right]}_{E_{\mathrm{nonbonded}}}$$

$$\boxed{E_{\mathrm{system}} = E_{\mathrm{bonded}} + E_{\mathrm{nonbonded}}}$$

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j)$$

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$$

Density Functional Theory (QM)



DFT

- ground state electron density($\rho(\mathbf{r})$) alone determines the system behavior
- overcomes many-body problem

Quickstep Method - DFT GPW [1]

- Uses both Gaussian and Planewaves to speedup calculation
- **Gaussian**: Localized basis set centered on atomic positions – Suitable for molecules
- **Planewave**: Best suited for condensed matter, solids, periodic system

$$\mathrm{E}[\rho(\mathbf{r})] = \mathrm{E}^{\mathrm{T}}[\chi(\mathbf{r})] + \mathrm{E}^{\mathrm{ne}}[\rho(\mathbf{r})] + \mathrm{E}^{\mathrm{ee}}[\rho(\mathbf{r})] + \mathrm{E}^{\mathrm{nn}} + \mathrm{E}^{\mathrm{XC}}[\rho(\mathbf{r})]$$

K.E electrons	nuclear-electron	electron-electron	nuclear-nuclear	XC
Gaussian	Gaussian+PW	PW	PW	PW

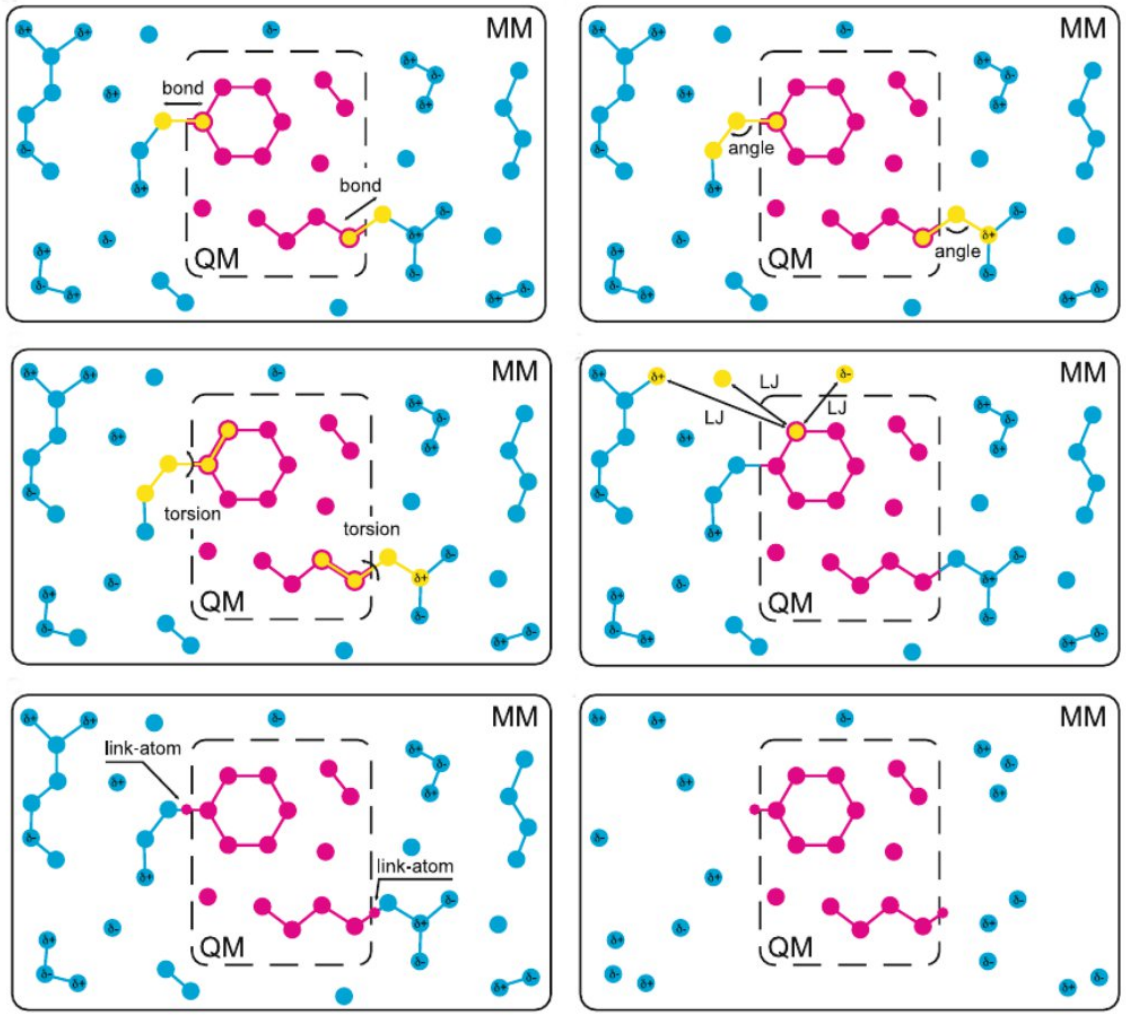
Additive QM/MM

- $E(\mathbf{r}_\alpha, \mathbf{r_a}) = E^{QM}(\mathbf{r}_\alpha) + E^{MM}(\mathbf{r_a}) + E^{QM/MM}(\mathbf{r}_\alpha, \mathbf{r_a})$

- Mechanical embedding: Electronic wavefunction isolated
  - QM-MM interactions defined at MM level

- Electrostatic Embedding: MM charges as one electron operators
  - MM polarize QM, but not vice versa

- Polarization Embedding: QM ⇔ MM polarization
  - Computationally expensive

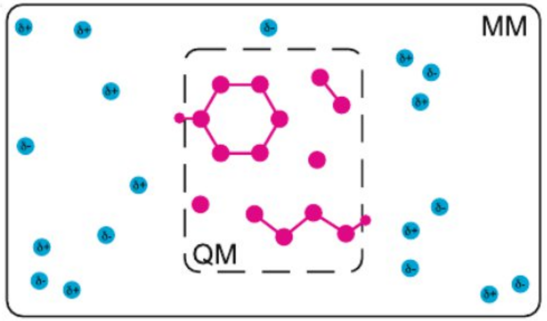


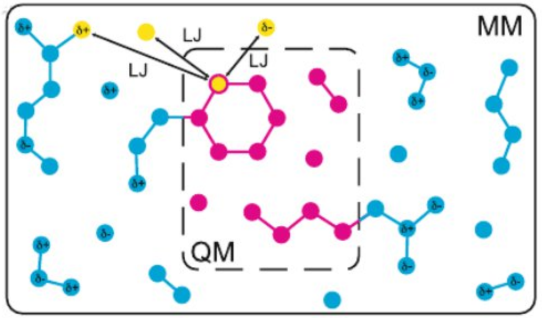
Electrostatic Embedding

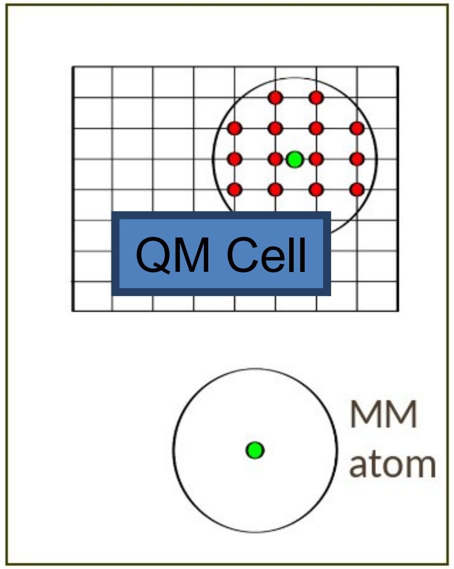
charge density
electron + nuclei

- $$\mathrm{E}_{\mathrm{NB}}^{\mathrm{QM/MM}}(\mathbf{r}_{\mathrm{i}}, \mathbf{R}_{\mathrm{I}}) = \frac{1}{4\pi\epsilon_0} \sum_{\mathrm{i}\in\mathrm{MM}} q_{\mathrm{i}} \int \frac{\rho}{|\mathbf{r}-\mathbf{r}_{\mathrm{i}}|} \mathrm{d}\mathbf{r} + \sum_{\substack{\mathrm{i}\in\mathrm{MM} \\ \mathrm{I}\in\mathrm{QM}}} \nu_{\mathrm{vdW}}(\mathbf{r}_{\mathrm{i}}, \mathbf{R}_{\mathrm{I}})$$
- MM atoms polarize the QM system
- charges of MM atoms and LJ parameters need to be parameterized
- electron-spill: electrons of QM atom close to MM atom gets trapped
- electrostatic embedding: GEEP
- MM Charge density in Gaussian

$$\frac{1}{(\sqrt{\pi}\, r_{c,a})^3} \exp\left(-\left(\frac{|\mathbf{r}-\mathbf{r}_{\mathrm{i}}|}{r_{c,a}}\right)^2\right)$$



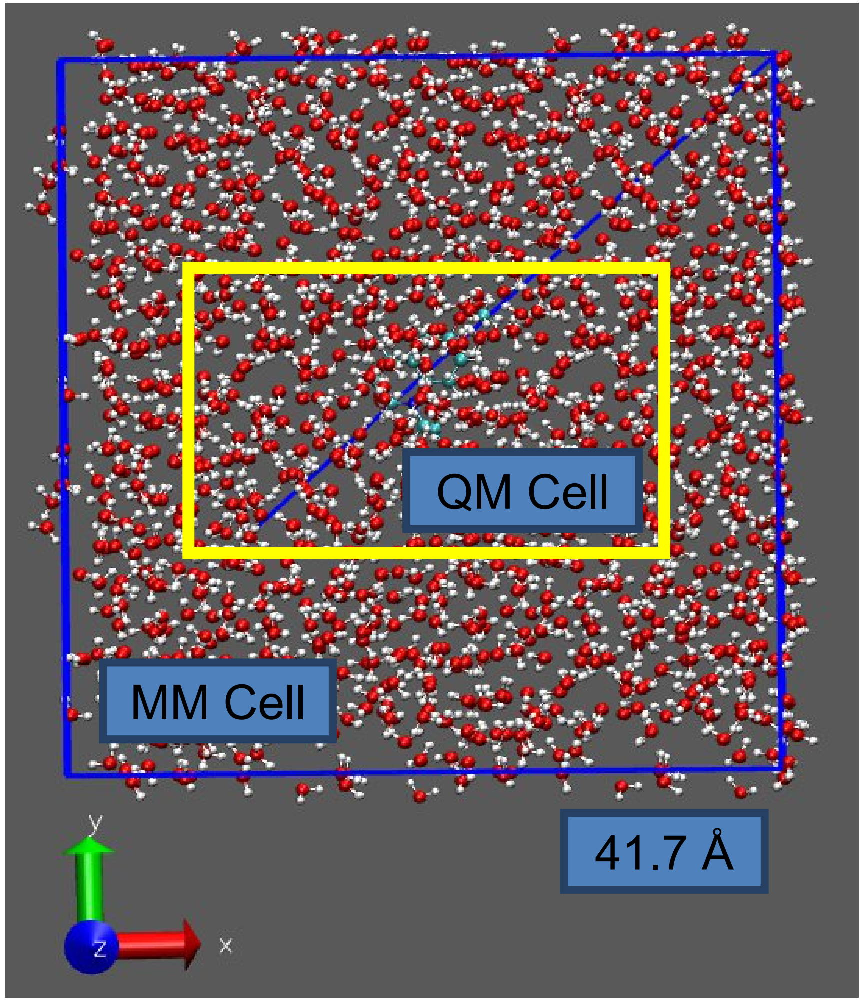


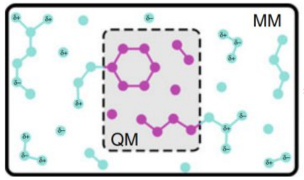


Gerrit Groenhof. Introduction to QM/MM Simulations, Humana Press, Totowa, NJ, 2013.

[GEEP: J. Chem. Th. Comp., 1, pp. 1176-1184 (2005)

Example1: Overview





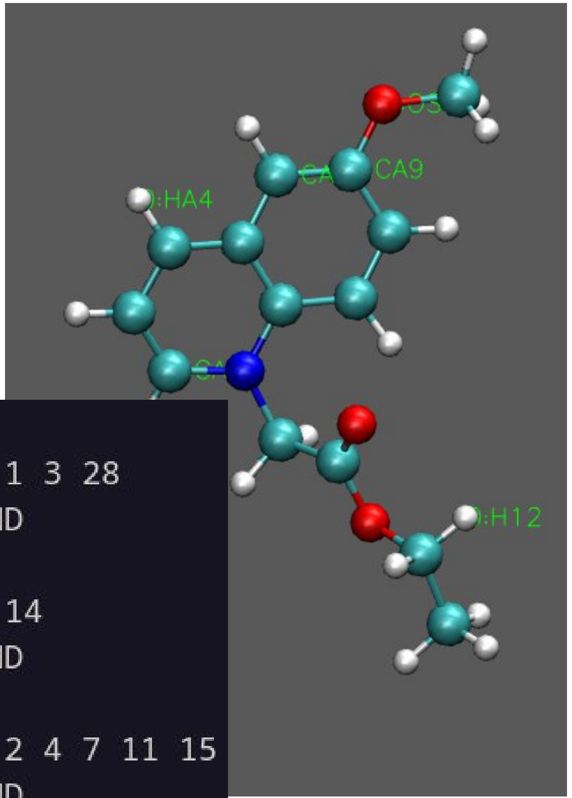

```
&QMMM  ! This defines the QS cell i
  ECOUPL GAUSS
  USE_GEEP_LIB 15
  &CELL
    ABC 13.79565 17.3825 10.76061
    PERIODIC XYZ
  &END CELL
  &PERIODIC  # apply periodic poten
```

```
&SUBSYS
  &CELL
    ABC 41.751 41.751 41.751
    ALPHA_BETA_GAMMA 90 90 90
    PERIODIC XYZ
  &END CELL
```

```
&KIND H
  BASIS_SET DZVP-MOLOPT-GTH
  ELEMENT H
  POTENTIAL GTH-BLYP-q1
&END KIND
&KIND C
  BASIS_SET DZVP-MOLOPT-GTH
  ELEMENT C
  POTENTIAL GTH-BLYP-q4
&END KIND
```

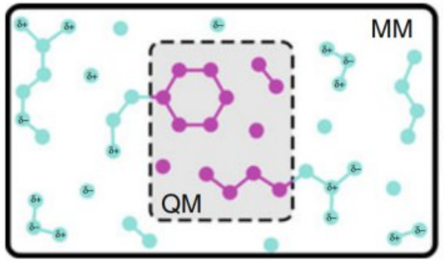
```
ATOM      8 HC          17.593  24.836  20.763
ATOM      9 HC          19.204  25.485  21.124
ATOM     10 HC          18.086  25.028  22.422
ATOM     11 C32         20.744  21.568  18.721
ATOM     12 H13         21.474  22.379  18.609
ATOM     13 H13         19.956  21.768  18.003
ATOM     14 NA          21.342  20.240  18.435
```

N-(6-methoxyquinolyl) acetoethyl ester




```
&QM_KIND O
  MM_INDEX 1 3 28
&END QM_KIND
&QM_KIND N
  MM_INDEX 14
&END QM_KIND
&QM_KIND C
  MM_INDEX 2 4 7 11 15
&END QM_KIND
&QM_KIND H
```

QMMM Inputfile



DFT Section

```
&FORCE_EVAL
  METHOD QMMM
  &DFT
    BASIS_SET_FILE_NAME BASIS_MOLOPT
    CHARGE 1
    MULTIPLICITY 1
    POTENTIAL_FILE_NAME POTENTIAL
    &MGRID
      COMMENSURATE
      CUTOFF 400
    &END MGRID
    &QS
      EPS_DEFAULT 1.0E-12
      METHOD GPW
    &END QS
    &SCF
      EPS_SCF 1.0E-6
      MAX_SCF 300
      &OT T
        MINIMIZER DIIS
        PRECONDITIONER FULL_ALL
        STEPSIZE 1.4999999999999999E-01
      &END OT
    &END SCF
    &XC
      &XC_FUNCTIONAL BLYP
      &END XC_FUNCTIONAL
    &END XC
  &END DFT
  &MM
    &FORCEFIELD
```

MM Section

```
  &END DFT
  &MM
    &FORCEFIELD
      EI_SCALE14 1.0
      PARMTYPE AMBER
      PARM_FILE_NAME MQAE.prmtop
      VDW_SCALE14 1.0
      &SPLINE
        EMAX_SPLINE 1.0E14
        RCUT_NB [angstrom] 12
      &END SPLINE
    &END FORCEFIELD
    &POISSON
      &EWALD
        ALPHA .40
        EWALD_TYPE SPME
        GMAX 80
      &END EWALD
    &END POISSON
  &END MM
  &QMMM  ! This defines the QS cell in t
    ECOUPL GAUSS
    USE_GEEP_LIB 15
    &CELL
      ABC 13.79565 17.3825 10.76061
      PERIODIC XYZ
    &END CELL
    &PERIODIC  # apply periodic potentia
      #turn on/off coupling/recoupling o
      &MULTIPOLE ON
      &END MULTIPOLE
```

QMMM

QM Cell

QM atoms

```
    &END POISSON
  &END MM
  &QMMM  ! This defines the QS cell i
    ECOUPL GAUSS
    USE_GEEP_LIB 15
    &CELL
      ABC 13.79565 17.3825 10.76061
      PERIODIC XYZ
    &END CELL
    &PERIODIC  # apply periodic poten
      #turn on/off coupling/recouplin
      &MULTIPOLE ON
      &END MULTIPOLE
    &END PERIODIC
    &QM_KIND O
      MM_INDEX 1 3 28
    &END QM_KIND
    &QM_KIND N
      MM_INDEX 14
    &END QM_KIND
    &QM_KIND C
      MM_INDEX 2 4 7 11 15 17 19 21 2
    &END QM_KIND
    &QM_KIND H
      MM_INDEX 5 6 8 9 10 12 13 16 18
    &END QM_KIND
  &END QMMM
  &SUBSYS
    &CELL
      ABC 41.751 41.751 41.751
```

Total Cell

BASIS SET

```
  &SUBSYS
    &CELL
      ABC 41.751 41.751 41.751
      ALPHA_BETA_GAMMA 90 90 90
      PERIODIC XYZ
    &END CELL
    &KIND H
      BASIS_SET DZVP-MOLOPT-GTH
      ELEMENT H
      POTENTIAL GTH-BLYP-q1
    &END KIND
    &KIND C
```

FF Data

```
      POTENTIAL GTH-BLYP-q5
    &END KIND
    &TOPOLOGY
      CONN_FILE_FORMAT AMBER
      CONN_FILE_NAME MQAE.prmtop
      COORD_FILE_FORMAT PDB
      COORD_FILE_NAME MQAE.pdb
    &END TOPOLOGY
  &END SUBSYS
&END FORCE_EVAL
```

Topology

`CONN_FILE_FORMAT`: *enum = GENERATE*

Aliases: CONNECTIVITY

Usage: *CONN_FILE_FORMAT (PSF|UPSF|MOL_SET|GENERATE|OFF|G87|G96|AMBER|USER)*

Valid values:

- `PSF` Use a PSF file to determine the connectivity. (support standard CHARMM/XPLOR and EXT CHARMM)
- `UPSF` Read a PSF file in an unformatted way (useful for not so standard PSF).
- `MOL_SET` Use multiple PSF (for now...) files to generate the whole system.
- `GENERATE` Use a simple distance criteria. (Look at keyword BONDPARM)
- `OFF` Do not generate molecules. (e.g. for QS or ill defined systems)
- `G87` Use GROMOS G87 topology file.
- `G96` Use GROMOS G96 topology file.
- `AMBER` Use AMBER topology file for reading connectivity (compatible starting from AMBER V.7)
- `USER` Allows the definition of molecules and residues based on the 5th and 6th column of the COORD section. This option can be handy for the definition of molecules with QS or to save memory in the case of very large systems (use PARA_RES off).

Ways to determine and generate a molecules. Default is to use GENERATE [Edit on GitHub]

`COORD_FILE_FORMAT`: *enum = OFF*

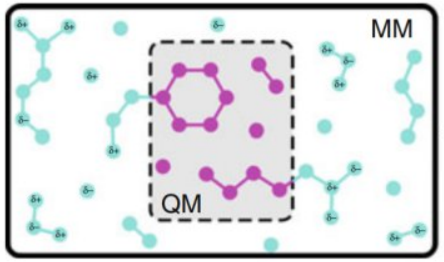
Aliases: COORDINATE

Usage: *COORD_FILE_FORMAT (OFF|PDB|XYZ|G96|CRD|CIF|XTL|CP2K)*

Valid values:

- `OFF` Coordinates read in the &COORD section of the input file
- `PDB` Coordinates provided through a PDB file format
- `XYZ` Coordinates provided through an XYZ file format
- `G96` Coordinates provided through a GROMOS96 file format
- `CRD` Coordinates provided through an AMBER file format
- `CIF` Coordinates provided through a CIF (Crystallographic Information File) file format
- `XTL` Coordinates provided through a XTL (MSI native) file format
- `CP2K` Read the coordinates in CP2K &COORD section format from an external file. NOTE: This file will be overwritten with the latest coordinates.

Set up the way in which coordinates will be read. [Edit on GitHub]



Inputfile

```
      POTENTIAL GTH-BLYP-q5
    &END KIND
    &TOPOLOGY
      CONN_FILE_FORMAT AMBER
      CONN_FILE_NAME MQAE.prmtop
      COORD_FILE_FORMAT PDB
      COORD_FILE_NAME MQAE.pdb
    &END TOPOLOGY
  &END SUBSYS
&END FORCE_EVAL
```

Example 2

P Peak Scenario

- Peak P scenario could happen soon
- Oil Crisis in 1970; N0 alternative for P